BLUE LOTUS RESEARCH INTELLIGENCE

Quantum Computing Inc

Photonic Thin Film Chips + CUDA-Q Datacentre Acceleration — Small-Cap High-Risk

PhD Due Diligence Report | BLRI-DD-QUBT-AUG2026

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Six-Method Framework: Quantitative · Qualitative · Ethnographic · Superforecasting · Adversarial-Collaborative · Seldon Thesis

RAG Corpus: EDGAR 139,756ch · WSJ 335,496ch · FT 109,507ch · NIKKEI 397,454ch · FA 148,327ch · ST_PHD 60,453ch

PART I — SELDON THESIS AXIS

Quantum Photonic Integration Units for AI Datacentres

Quantum Computing Inc (QCI) is pursuing a distinct strategy from hardware quantum computers: it builds Quantum Photonic Integration Units (QPIUs) — silicon photonic chips that accelerate AI inference in datacentres by reducing power consumption via optical matrix multiplication. This is not quantum computing in the traditional sense — it is quantum-inspired photonic acceleration for classical AI workloads. In Seldon K-variable terms: QPIUs are a near-term ξ variable play — they could reduce energy consumption per AI inference by 10-100×, broadening AI accessibility.

QCI also offers CUDA-Q compatible quantum circuit software tools targeting NVIDIA's GPU+quantum hybrid computing stack. This software product gives QCI exposure to NVIDIA's 90% AI accelerator market share — a distribution channel of enormous potential reach if CUDA-Q adoption accelerates.

Source: moomoo OpenD [2026-08-13] · QCI investor relations · NVIDIA CUDA-Q programme

PART II — QUALITATIVE ANALYSIS

Thin Film Lithium Niobate (TFLN): Photonic Chip Advantage

QCI's photonic chip technology is based on Thin Film Lithium Niobate — a material with exceptional electro-optic properties enabling ultra-fast, low-power optical modulation. TFLN photonic chips are used in telecommunications infrastructure today; QCI is extending them into AI datacentre acceleration. This is a commercialisation bet on an existing material technology, reducing fundamental physics risk vs pure quantum hardware companies.

CUDA-Q Integration: NVIDIA Distribution Channel

NVIDIA's CUDA-Q platform enables hybrid classical-quantum programming. QCI's compatibility with CUDA-Q positions it within NVIDIA's AI ecosystem — the largest software distribution channel in compute. If NVIDIA customers adopt CUDA-Q for quantum workloads, QCI's tools gain automatic visibility to hundreds of enterprise AI teams.

Source: QCI 10-Q EDGAR · NVIDIA CUDA-Q documentation · TFLN photonic literature

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PART III — QUANTITATIVE ANALYSIS

VALUATION SNAPSHOT [moomoo OpenD 2026-08-13]
Price: $9.065 | Prev Close: $8.92 | Mkt Cap: $2.052B
EPS TTM: -$0.11 | Net Income: -$24.9M | Div: 0%
52w: $6.18–$25.84 | Shares: 226.3M

Metric	Value	Source
Price (2026-08-13)	$9.065	moomoo OpenD
Market Cap	$2.052B	moomoo OpenD
EPS TTM	-$0.11	moomoo OpenD
Net Income TTM	-$24.9M	moomoo OpenD
52-Week Range	$6.18–$25.84	moomoo OpenD
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PART IV — COMPETITIVE LANDSCAPE

Competitor	Type	Mkt Cap	vs QUBT
Lightmatter (private)	Photonic AI accelerator	Private >$1B	Direct competitor; QUBT listed
Luminous Computing (private)	Photonic AI	Private	Direct; QUBT CUDA-Q edge
NVIDIA (not listed as quantum)	GPU AI compute	$3.5T	Partner channel; CUDA-Q synergy
IonQ (IONQ)	Quantum hardware	$17.751B	Different market — QUBT is photonic AI, not QC

PART V — ETHNOGRAPHIC PORTRAITS

FINDING F-5-A: Portrait 1: The AI Datacentre Power Engineer (COMPOSITE)

A power infrastructure engineer at a Tier 4 datacentre evaluates photonic AI accelerators to reduce PUE from 1.4 to 1.1. 'If QCI's QPIU cuts inference power by 30%, that's $5M/year in power savings at our scale. We'd pay $2M for that.' Represents the datacentre energy efficiency market where QCI's photonic chips have immediate commercial value if performance claims are validated.

COMPOSITE / ILLUSTRATIVE — drawn from datacentre energy efficiency patterns

FINDING F-5-B: Portrait 2: The NVIDIA Developer Using CUDA-Q (COMPOSITE)

An AI research engineer at a pharmaceutical company uses NVIDIA CUDA-Q for hybrid quantum-classical molecular simulation. He discovered QCI's quantum circuit tools through the CUDA-Q ecosystem. 'QCI tools integrate cleanly with my existing GPU workflow. I don't need to learn a new SDK.' Represents the NVIDIA developer community that QCI targets via CUDA-Q distribution.

COMPOSITE / ILLUSTRATIVE — drawn from NVIDIA CUDA-Q developer community patterns

FINDING F-5-C: Portrait 3: The Small-Cap Quantum Investor (COMPOSITE)

A retail investor in quantum computing ETFs notices QUBT in her QTUM ETF holdings. 'QCI is tiny — $2B market cap, $24M net loss. But if their photonic chip works in datacentres, this is a 10× from here.' Represents the speculative retail investor segment that drives QUBT's volatile price action — the stock's 52-week range ($6.18–$25.84) reflects this speculative character.

COMPOSITE / ILLUSTRATIVE — drawn from small-cap quantum retail investor patterns

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PART VI — ADVERSARIAL COLLABORATION (5 HYPOTHESES)

HH1: HH1: Lightmatter or Luminous achieves photonic AI commercialisation first

EVIDENCE FOR:
Both companies have $500M+ in venture funding; dedicated teams; if either achieves hyperscaler adoption, QUBT becomes irrelevant

EVIDENCE AGAINST:
QUBT is the only listed company in this space — capital access advantage; CUDA-Q partnership gives QUBT distribution Lightmatter lacks

QUANTITATIVE TEST
P(private photonic AI competitor commercialises before QUBT by 2027)=40%

VERDICT: BEAR — private competition is the primary near-term risk
Implication: Monitor Lightmatter/Luminous funding rounds and partnership announcements

HH2: HH2: NVIDIA acquires QUBT for CUDA-Q quantum integration

EVIDENCE FOR:
NVIDIA's quantum computing strategy (CUDA-Q) needs hardware; QUBT's photonic tools + TFLN expertise is acquirable at $2B; NVIDIA can afford any acquisition

EVIDENCE AGAINST:
NVIDIA prefers partnerships to acquisitions in quantum; QUBT's technology is early-stage; NVIDIA may wait for more mature assets

QUANTITATIVE TEST
P(NVIDIA acquisition at >50% premium by 2027)=10%

VERDICT: BULL scenario — acquisition at $13-15/share
Implication: Monitor NVIDIA's quantum computing M&A activity and CUDA-Q roadmap

HH3: HH3: QPIU performance claims are overstated; validation fails

EVIDENCE FOR:
Photonic AI accelerator benchmarks are often tested in optimal conditions; real datacentre deployment may show 5-10× not 100× improvement; customers demand independent validation before purchase

EVIDENCE AGAINST:
TFLN photonic modulation has established physics basis; QCI has demonstrated working prototypes in controlled conditions

QUANTITATIVE TEST
P(QPIU independent validation shows <5× improvement, triggering selloff)=30%

VERDICT: BEAR — overstated performance claims would be fatal at $2B valuation
Implication: Demand independent third-party performance benchmarks before full position

HH4: HH4: Datacentre energy crisis accelerates QPIU adoption

EVIDENCE FOR:
AI training clusters consuming 1-2 GW each; US grid capacity limits new datacenter builds; photonic inference acceleration that reduces power consumption by 50%+ is urgently needed

EVIDENCE AGAINST:
QPIU adoption requires integration into existing CUDA workflows; transition from GPU to photonic takes 2-3 year procurement cycles

QUANTITATIVE TEST
P(QPIU commercial deployments in 3+ major datacentres by 2027)=20%

VERDICT: BULL — energy crisis as QUBT adoption accelerant is the strongest bull thesis
Implication: Monitor power usage effectiveness (PUE) mandates and datacentre energy regulations

HH5: HH5: QUBT dilutes heavily and stock returns to $6

EVIDENCE FOR:
Net loss -$24.9M; revenue near zero; QPIU commercial timeline uncertain; equity raises below $9 price destroy shareholder value

EVIDENCE AGAINST:
Smaller absolute burn than hardware companies; TFLN chip manufacturing cost is manageable vs cryogenic system development

QUANTITATIVE TEST
P(dilutive raise at <$7/share within 18 months)=35%

VERDICT: BEAR — small cap with pre-revenue product is highest dilution risk
Implication: Monitor cash runway; any QPIU commercial contract announcement before raise

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PART VII — SUPERFORECASTING (10 SCENARIOS)

SCENARIO 1: First QPIU commercial datacentre deployment (P = 18%%)

QCI deploys QPIU in 1-3 commercial datacentre customers; performance validated at >10× power reduction; first revenue recognition; stock $15-18

Driver: Customer announcements; datacentre partnership press releases

Falsifier: QPIU delayed past 2027; performance below claims in customer pilot

SCENARIO 2: NVIDIA CUDA-Q integration drives developer adoption (P = 30%%)

QCI software tools adopted by 5,000+ CUDA-Q developers; brand recognition in quantum-adjacent ML community; stock holds $9-12 range on developer momentum

Driver: CUDA-Q developer survey data; QCI download counts

Falsifier: NVIDIA builds competing quantum tools in-house; QCI SDK marginalised

SCENARIO 3: Base case: pre-revenue, survival mode (P = 35%%)

QCI maintains CUDA-Q positioning; QPIU pilot customers; losses narrow slightly; stock $7-11; equity raise in 2027

Driver: Quarterly R&D milestones; pilot customer announcements

Falsifier: No pilot customers by end 2026; dilutive raise triggers selloff

SCENARIO 4: Acquisition by photonic infrastructure company (P = 10%%)

Lumentum, II-VI, or Intel acquires QCI for TFLN photonic IP; acquisition at $12-15/share; 30-65% premium

Driver: Photonic component company M&A activity; TFLN IP licensing

Falsifier: Acquirers build TFLN capability internally; QCI not strategically relevant

SCENARIO 5: Quantum winter + pre-revenue status destroys value (P = 20%%)

Quantum sector selloff; QCI has no revenue to defend valuation; stock falls below 52w low $6.18; possible survival risk

Driver: Quantum ETF outflows; sector sentiment collapse

Falsifier: QPIU commercial milestone announced; first contract signed

SCENARIO 6: Datacentre energy mandate creates urgent demand (P = 15%%)

US EPA or EU efficiency mandate requires 30% PUE improvement; QPIU is only commercial photonic solution available; emergency procurement drives 2-3 early-adopter contracts

Driver: EPA/EU energy efficiency regulations for datacentres

Falsifier: Regulation delayed; GPU efficiency improvements eliminate urgency

SCENARIO 7: Partnership with AWS or Azure for quantum inference cloud (P = 8%%)

AWS or Azure integrates QPIU into inference cloud services; quantum photonic inference available as managed service; stock re-rates to $18-20 on hyperscaler partnership

Driver: AWS/Azure product roadmap leaks; hyperscaler quantum announcements

Falsifier: Hyperscalers build own photonic inference capability via proprietary hardware

SCENARIO 8: TFLN technology licensed to telecom (5G/6G) (P = 12%%)

Nokia, Ericsson, or Qualcomm licenses QCI's TFLN photonic modulator for 5G/6G; licensing revenue provides non-dilutive cash; stock $10-13 on licensing model

Driver: Telecom partnership announcements; 6G photonic integration specs

Falsifier: Telecom builds proprietary TFLN capability; no licensing opportunity

SCENARIO 9: QUBT included in quantum computing index / ETF rebalancing (P = 20%%)

Major index provider adds QUBT to quantum computing index; ETF buying creates 15-20% price support; institutional coverage initiates

Driver: Index provider quantum computing sector definition updates

Falsifier: QUBT excluded from major indices due to small float/liquidity

SCENARIO 10: Bull case: QPIU becomes the GPU for photonic AI by 2030 (P = 4%%)

QCI's QPIU achieves commercial dominance in photonic inference; $100M+ revenue by 2029; stock $40+ ($9B market cap); profitability visible

Driver: Multiple datacentre deployments; major partnership announcements; performance validated

Falsifier: All competitor photonic AI companies deploy before QCI; technology obsoleted

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PART VIII — SELDON VARIABLE DEEP DIVE

Variable	QUBT Impact	Direction	Magnitude
K	CUDA-Q integration enables hybrid quantum-classical K acceleration	↑ Mild	K+ marginal
ξ	QPIU power reduction democratises AI inference — broadens access	↑ Moderate	ξ+ if commercialised
Ω	No cryptographic threat; photonic AI not Shor-capable	↓ Reduces	Ω-minus relative to QC
I	Minimal institutional impact at current scale	→ Neutral	I neutral

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PART IX — RISK REGISTER

Risk	Probability	Impact	Mitigation
Performance overstatement	HIGH (30%)	CRITICAL	Demand independent third-party validation
Private competitor pre-emption	HIGH (40%)	HIGH	CUDA-Q moat; listed capital access
Dilution at low price	HIGH (35%)	HIGH	Monitor cash quarterly; pilot customer deadline
Quantum winter sector selloff	MEDIUM (20%)	VERY HIGH	No revenue floor — highest volatility risk

Technology pivot risk (pre-revenue)	MEDIUM (20%)	MEDIUM	Management track record review

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CONCLUSION — 6-METHOD SYNTHESIS VERDICT

CONVICTION VERDICT: HIGHLY SPECULATIVE — OPTION VALUE ON PHOTONIC AI, NOT PURE QUANTUM

QUBT at $9.065 (mkt cap $2.052B) is the highest-risk report in this quantum series. It is not a pure quantum computing company — it is a photonic AI accelerator startup with quantum software tools. The investment thesis: QPIU solves the datacentre energy crisis; CUDA-Q integration provides NVIDIA distribution. Key risks: performance overstatement, private competition, pre-revenue dilution. Seldon alignment: ξ+ if commercialised (AI democratisation); minimal K impact. HUMAN_ONLY_MANUAL. All figures: moomoo OpenD [2026-08-13]. RAG Corpus: EDGAR 139,756ch · WSJ 335,496ch · FT 109,507ch · NIKKEI 397,454ch · FA 148,327ch · ST_PHD 60,492ch.